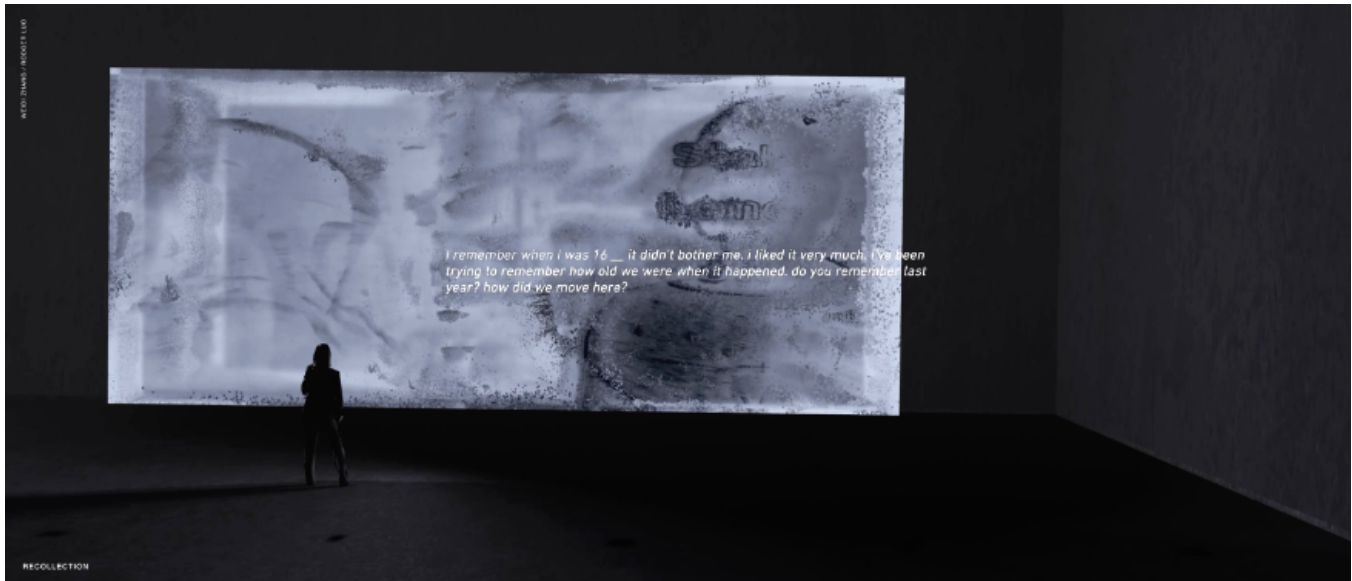


Re:collection

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1 PROJECT URL

<http://www.zhangweidi.com/recollection>

2 DESCRIPTION OF ARTWORK

Re:collection is an interactive AI art installation that assembles synthetic collective visual memories based on language input, blurring the boundaries between remembrance and imagination through intelligent system design and experimental visualization.

According to Dr. Mary Steedly, memories are a "densely layered, sometimes conflictual negotiation with the passage of time." As a result of Alzheimer's disease and related dementias, at least 50 million people suffer memory regression and cognitive difficulties. Meanwhile, text-to-image artificial intelligence (AI) systems have gained a great deal of attention in 2022 as exciting forms of AI that generate visuals based on language input, which creates new possibilities for reassembling fragmented memories.

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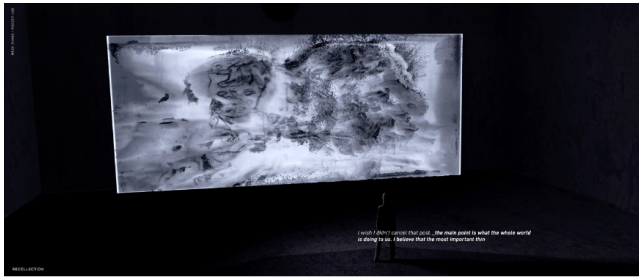
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- (1) When we coexist with machines, will we accumulate synthetic recollections of symbiotic imagination?
- (2) Is language capable of triggering and synthesizing memories?
- (3) Can we preserve collections of memories in the time capsule for cultural reproduction and re-imagination?
- (4) How does our collective memory inspire future new visual forms, aesthetics, and alternative narratives?

Through conditioning a pre-trained transformer-based AI model to learn Alzheimer patients' memories and descriptions, this customized AI application will facilitate multiple AI techniques, like speech recognition, text auto-completion, and text-to-image, to convert language input into image sequences of new memories. As an interactive experience, participants will describe their memories with fragmented sentences, and our system will automatically fill in details, creating new visual memories. By providing a conceptual framework for non-linear narratives, which constitute symbiotic imaginations and future scenarios of memories, culture production, and reproductions.

2.1 User Interaction

In the art installation, a participant will whisper fragmented memories into the microphone. The AI system will automatically fill in the details of the spoken words to complete the text with a narrative using a large language model. The completed narrative will be sent to a generative-AI model to generate synthetic images representing the memories based on the machine's interpretation. The images output by machines is further developed and visualized algorithmically as



an evolving interactive experience in print-making aesthetics. Technical Methodology This project integrates multiple AI techniques (speech recognition, text auto-completion, text-to-image model) with experimental data visualization to create a personalized experience that emphasizes computational aesthetics, personalized art experience, and human-machine co-creation. Three artistic and technical goals are outlined in it, including:

- Developing a prototype that blurs the line between imagining the future and remembering the past by integrating artificial intelligence system design and experimental visualization.
- Develop a conceptual framework for non-linear narratives that constitute symbiotic imaginations and offer alternative possibilities for the cure of memory regression.
- Combine automation with artistic originality to create a non-linear narrative that leads to the new abstraction of empathy.

3 ARTIST INFORMATION

Weidi Zhang is a Los Angeles and Phoenix-based new media artist / designer and an assistant professor of immersive experience design at Arizona State University (US). Her interdisciplinary art and design research investigates A Speculative Assemblage at the intersection of immersive media design, experimental data visualization, and interactive AI art. Her works are featured in international awards, such as Red Dot Design Award, Best In Show Awards in SIGGRAPH, Honorary Mention in Prix Ars Electronica, Juried Selection in Japan Media Arts Festival, Lumen Prize shortlists, and others. Her works have been exhibited at international venues such as ISEA, Times Art Museum, Society For Arts and Technology [SAT], SwissNex Gallery, SIGGRAPH & SIGGRAPH ASIA, CVPR, IEEE VISAP, Zeiss-Planetarium Jena, Planetarium 1, and others. She holds her Ph.D. degree in Media Arts and Technology from the University of California, Santa Barbara, and an MFA degree in Art + Technology from the California Institute of the Arts.

Jieliang (Rodger) Luo is a Principal AI Research Scientist at the Autodesk AI Lab in San Francisco. He received his Ph.D. from UC Santa Barbara, where he explored the intersection of machine learning, robotics, and creativity. His current research focus is understanding how to learn complex behaviors with imitation and reinforcement learning and applying the insights for construction automation and architecture design. His work has been presented at SIGGRAPH, ICCV, ICRA, CoRL, IROS, ECCV, ICLR, and others.